

Jamere Golden

1. Attached file abc.xyz Download abc.xyz has been intentionally corrupted. Make every effort to repair the file. Describe the process you followed. Submit a snapshot of the result.

WinHex [2] - [abc(1).xyz]

File Edit Search Navigation View Tools Specialist Options Window Help

abc(1).xyz [unregistered]

File size: 54.4 KB
55,715 bytes

Default Edit Mode: modified

Undo level: 0

Undo reverses: keyboard input

Creation time: 03/27/2025 14:52:09

Last write time: 03/27/2025 14:52:10

Attributes: A

Icons: 0

Mode: hexadecimal

Offsets: hexadecimal

Bytes per page: 53x16=848

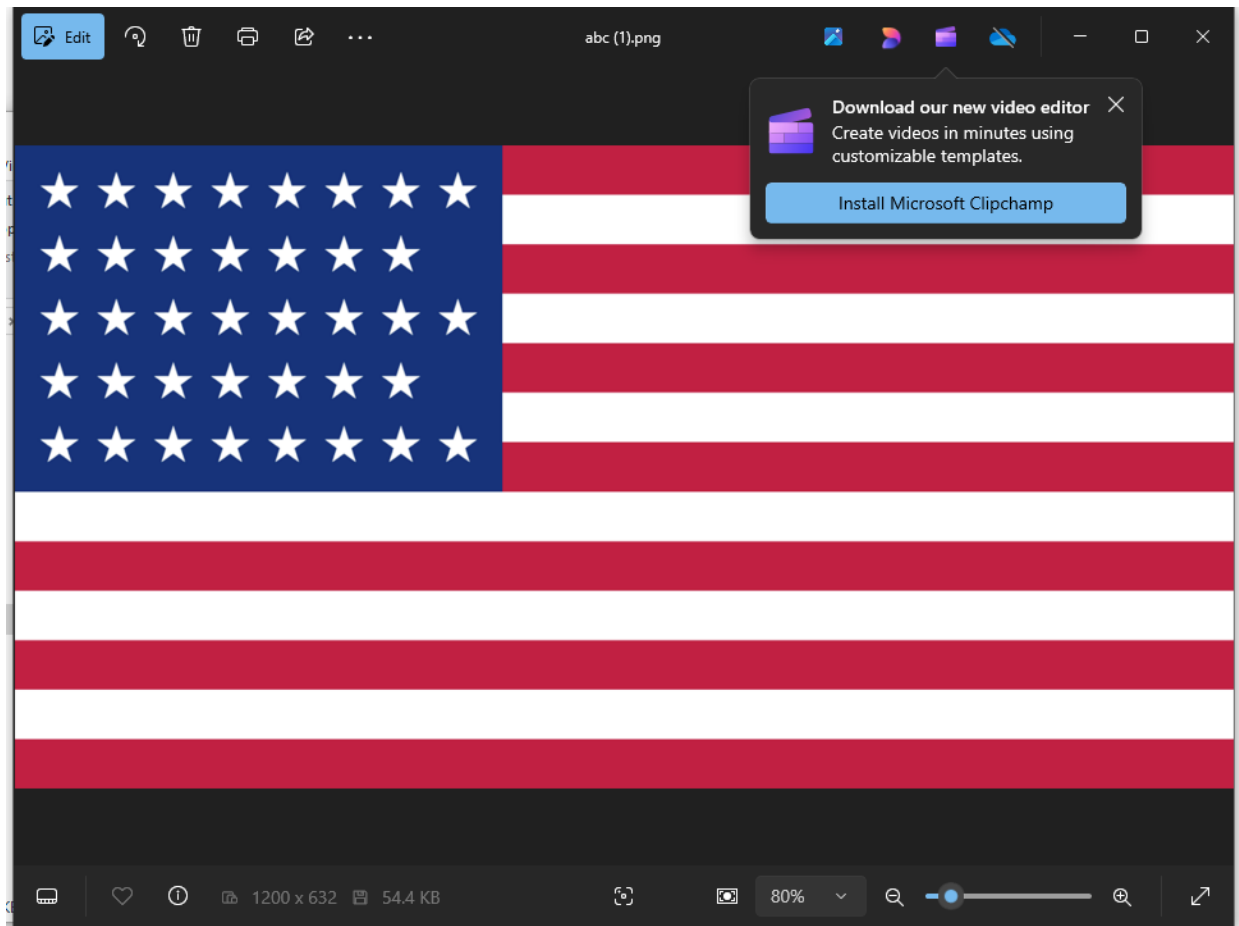
Window #: 1

Window count: 1

Clipboard: empty

Path for temp. files: 216 GB free
ers\STJA11~1\AppData\Local\Temp

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	ANSI ASCII
00000000	89	50	4E	47	0D	0A	1A	0A	00	00	00	0D	49	48	44	52	WPNQ IHDR
00000010	00	00	04	B0	00	00	02	78	08	06	00	00	00	75	4A	1A	° x uJ
00000020	2C	00	00	00	04	67	41	4D	41	00	00	D6	D8	D4	4F	58	, gAMA 000X
00000030	32	00	00	00	06	62	4B	47	44	00	FF	00	FF	00	FF	A0	2 bKGD y y y
00000040	BD	A7	93	00	00	00	09	70	48	59	73	00	00	00	48	00	h\$" pHYS H
00000050	00	00	48	00	46	C9	6B	3E	00	00	80	00	49	44	41	54	H FÉk> € IDAT
00000060	78	DA	EC	DD	77	78	5B	77	A1	3E	F0	F7	1C	6D	4B	B6	xÜiYwx[w; > ð÷ mKQ
00000070	65	59	96	E5	2D	EF	3D	63	3B	3B	CE	EA	48	5B	4A	CB	eY-â-i=c;;fiêH[JE
00000080	B8	5C	F6	B8	70	2F	F7	72	F9	B1	A1	EC	C2	85	42	E9	,\0,p/÷rù±;iâ.Bé
00000090	A2	B4	85	16	0A	A5	94	D1	9D	34	4D	9A	36	7B	2F	C7	c'... W"N 4M\$6(/ç
000000A0	8E	E3	0C	27	DE	23	F1	DE	DB	96	74	CE	EF	0F	39	6E	Žâ 'b#âbû-tîi 9n
000000B0	4C	E2	D4	5B	B2	78	3F	CF	C3	F3	50	5B	39	D1	AB	E5	LâÖ[+x?IâoP[9N«â
000000C0	7C	5F	7F	87	80	A4	2F	CA	20	22	22	A2	05	63	94	EC	l_ +eH/Ê ""c c'i
000000D0	78	AE	B1	14	46	C9	CE	07	83	88	88	88	88	68	0A	44	x0± FÉî f""h D
000000E0	3E	04	44	44	44	44	44	44	44	44	E4	C9	58	60	11	11	> DDDDDDDaEX`
000000F0	11	11	11	11	11	11	91	47	63	81	45	44	44	44	44	44	`Gc EDDDDD
00000100	44	44	44	1E	8D	05	16	11	11	11	11	11	11	11	11	79	DDD y
00000110	34	16	58	44	44	44	44	44	44	44	E4	D1	58	60	11	11	4 XDDDDDDDaNX`
00000120	11	11	11	11	11	11	11	91	47	63	81	45	44	44	44	44	`Gc EDDDD
00000130	44	44	44	44	1E	8D	05	16	11	11	11	11	11	11	11	11	DDDD
00000140	79	34	16	58	44	44	44	44	44	44	E4	D1	58	60	11	11	y4 XDDDDDDDaNX`
00000150	11	11	11	11	11	11	11	11	91	47	63	81	45	44	44	44	`Gc EDDDD
00000160	44	44	44	44	44	1E	8D	05	16	11	11	11	11	11	11	11	DDDD
00000170	11	79	34	16	58	44	44	44	44	44	E4	D1	58	60	11	11	y4 XDDDDDDDaNX`
00000180	60	11	11	11	11	11	11	11	11	91	47	63	81	45	44	44	`Gc EDD
00000190	44	44	44	44	44	44	1E	8D	05	16	11	11	11	11	11	11	DDDDDD
000001A0	11	11	79	34	16	58	44	44	44	44	E4	D1	58	60	11	11	y4 XDDDDDDDaNX`
000001B0	58	60	11	11	11	11	11	11	11	11	91	47	63	81	45	44	X`Gc ED
000001C0	44	44	44	44	44	44	44	1E	8D	05	16	11	11	11	11	11	DDDDDD
000001D0	11	11	11	79	34	16	58	44	44	44	E4	D1	58	60	11	11	y4 XDDDDDDDaNX`
000001E0	D1	58	60	11	11	11	11	11	11	11	11	91	47	63	81	45	ÑX`Gc E
000001F0	44	44	44	44	44	44	44	44	1E	8D	05	16	11	11	11	11	DDDDDDDD
00000200	11	11	11	11	79	B4	45	55	60	29	44	01	82	20	78	EF	y'EU'D , xî
00000210	93	C1	7C	8B	3E	9F	E8	CD	F9	04	01	A2	E8	BD	F9	04	"Á <>ÿëîù eè±ù
00000220	E6	5B	F4	F9	14	5E	9C	8F	88	88	88	88	E8	5F	DD	E2	æ[óù ^æ ""è yâ
00000230	29	B0	24	09	9B	0A	D2	10	13	61	06	24	D9	FB	9E	09	)°ç > Ò a SÜûž
00000240	A7	84	5B	56	24	23	29	26	18	90	24	AF	CC	57	90	97	\$„[V\$#)& \$TW -
00000250	80	8C	C4	30	C0	E9	9D	F9	96	67	C5	60	49	5A	94	D7	€ÇÅ0âé ù-gÅ`IZ"*
00000260	E6	5B	92	16	85	E5	59	31	5E	9B	2F	23	31	0C	05	79	æ[' ..ÅY1^>/#1 y
00000270	09	DE	99	4F	92	90	14	13	8C	5B	56	24	7B	69	3E	19	E"O' E[V\$[i
00000280	31	11	66	6C	2A	48	F3	CE	CF	4F	22	22	22	22	22	5A	1 fl*HóïïO""""Z
00000290	24	05	96	2C	C3	68	34	E0	53	77	2F	C3	C6	E5	49	50	\$ -,Äh4âSw/ÄÄâIP
000002A0	28	BC	6C	E5	A3	2C	C3	D7	CF	07	1F	BB	2B	1F	B7	AF	(!lâé,Ä*ï »+`-`
000002B0	4A	85	52	A9	F0	BA	7C	3E	7A	2D	3E	B2	69	09	EE	5A	J..R00° >z->±i iz
000002C0	9B	0E	95	5A	E9	75	F9	74	3A	35	3E	74	5B	0E	EE	D9	> *Zéuùt:5>t[iÜ
000002D0	90	09	8D	56	05	78	53	C7	2A	03	1A	AD	0A	F7	6C	C8	V xSç* - ÷iE
000002E0	C4	87	6E	CB	81	4E	A7	06	64	EF	2A	91	55	6A	25	EE	Ä+ñE N\$ di*'Uj1i
000002F0	5A	9B	8E	8F	6C	5A	02	1F	BD	D6	EB	F2	29	95	0A	DC	Z>ž lZ ÷0ë0)• Ü
00000300	BE	2A	15	1F	BB	2B	1F	BE	7E	3E	5E	97	4F	1A	10	B1	%* »+ %->^~O; ±
00000310	71	79	12	3E	75	F7	32	18	8D	06	AF	CB	47	44	44	44	qy >u÷2 EGDD
00000320	44	44	8B	A5	C0	72	4A	48	4F	08	45	6E	5A	2A	EE	2C	DDçVÄrJHO EnZ\$1,
00000330	48	73	0D	A0	BD	89	24	23	29	3A	18	CB	32	6D	D8	B4	Hs 'h\$#): ÈZm0'
00000340	26	15	BE	BE	3A	EF	1A	80	8D	CD	8E	58	B5	24	16	B7	& %4:1 € iZxu\$.



2. Attached file file0.txt Download file0.txt has been edited in WinHex. Two operations have been applied in no particular order; a one-bit shift in either the left or the right direction and an XOR application with a prime number in the range 1 to 10. Repair the file. Explain the process. Submit a snapshot of the file to canvas.

Here are my trials before I got it:

Downloads - Shortcut.Ink Downloads - Shortcut.Ink file0 (1).txt

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
00000000	E1	E8	FD	EB	B2	C6	C6	7F	E2	E7	E6	F7	E0	B2	D4	FD	ANSI ASCII
00000010	E0	F7	FC	E1	FB	F1	E3	3B	28	00	00						44y*EE 4qm+4*0y 4=0a0a0a; (

Modify Data

☐ Add ☐ hexadecimal

Integer type: Value range:
☐ stay within limits
☒ allow over/underflow

☐ Reverse byte order ☐ Invert bits

☐ Left shift by 1 bit ☐ XOR

☐ Right shift by 1 bit ☐ OR

☐ Shift by bytes ☐ AND

☐ Circular left rotation ☐ ROT13

dit Search Navigation View Tools Specialist Options Window Help

Downloads - Shortcut.Ink Downloads - Shortcut.Ink file0 (1).txt

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
00000000	E0	E9	FC	EA	B3	C7	C7	7E	E3	E6	E7	F6	E1	B3	D5	FC	44u4*CC-4eq04*0u
00000010	E1	F6	FD	E0	FA	F0	E2	3A	29	01	01						44y4u0a0a;)

Modify Data

☐ Add ☐ hexadecimal

Integer type: Value range:
☐ stay within limits
☒ allow over/underflow

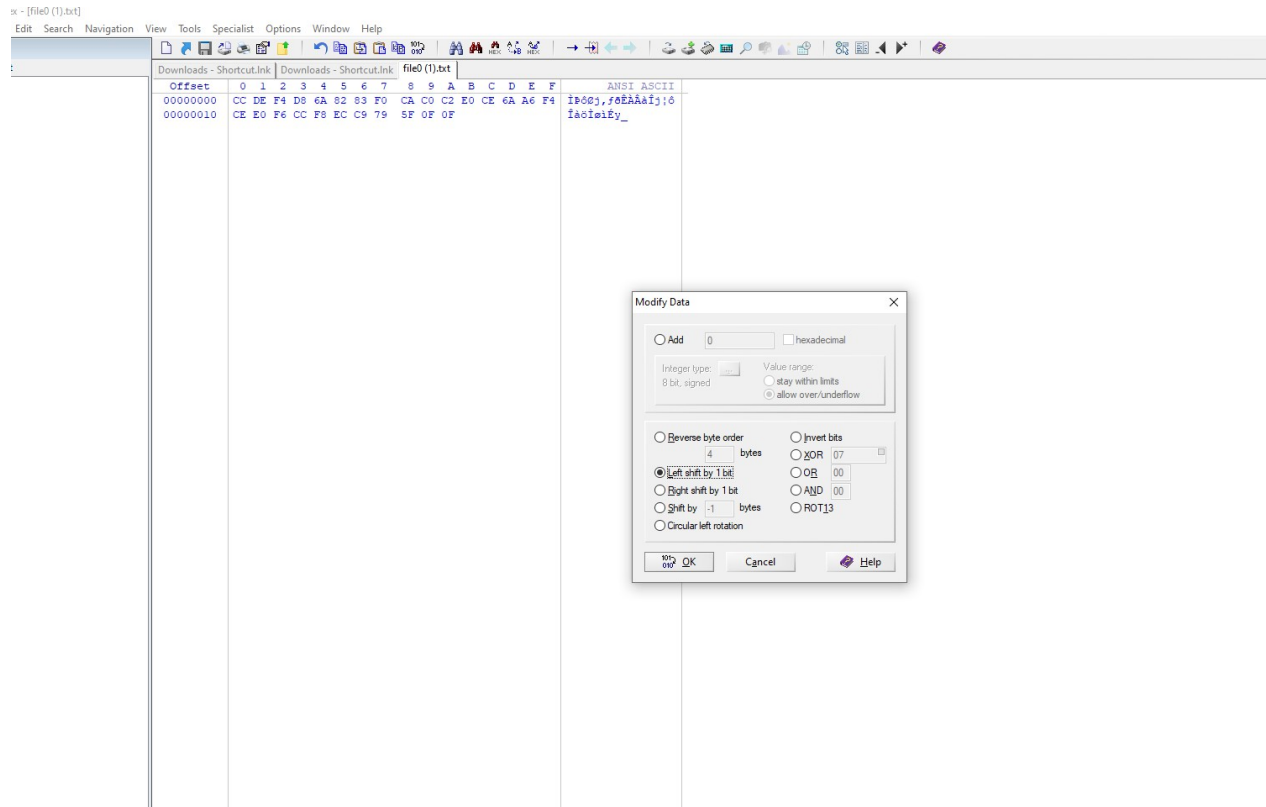
☐ Reverse byte order ☐ Invert bits

☐ Left shift by 1 bit ☒ XOR

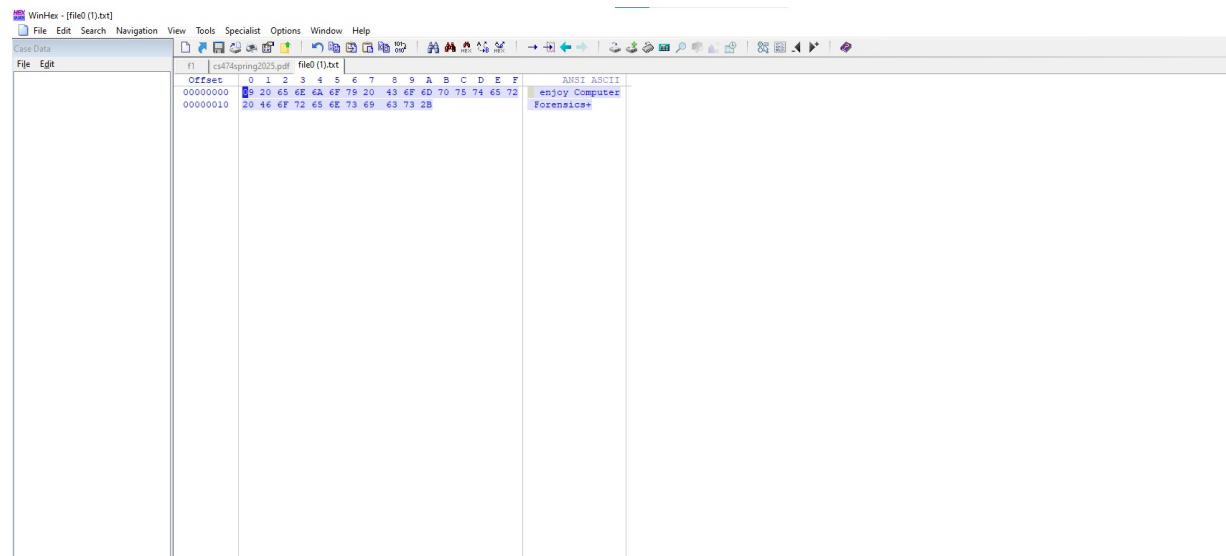
☐ Right shift by 1 bit ☐ OR

☐ Shift by bytes ☐ AND

☐ Circular left rotation ☐ ROT13



Here is the trial when I finally cracked the code:



3. In a hard drive formatted as NTFS, select a file of your choice that has a non-resident 0x80 data attribute. Report the following information. Use relevant snapshots and highlights to illustrate the process.

Hard drive geometry / Sectors/Cluster and Bytes/Sector

FF 47 92 3B 00 00 00 00

In little endian : 3B 92 47 FF = 999442431 , total sectors

Total cluster = sectors *8 = 7995539448

Cluster number : 786,432

```
0000000000 EB 52 90 4E 54 46 53 20-20 20 20 00 02 08 00 00 ER NTFS .....
0000000010 00 00 00 00 00 F8 00 00-3F 00 00 00 C8 0B 00 .....e-?g-E-
0000000020 00 00 00 00 80 00 80 00-47 92 3B 00 00 00 00 .....FC:
0000000030 00 00 0C 00 00 00 00 00-02 00 00 00 00 00 00 .....
0000000040 F6 00 00 00 01 00 00 00-20 3D C0 BE 88 C0 BE BC 0 .....ANAN
0000000050 00 00 00 00 FA 33 C0 8E-D0 BC 00 7C FB 68 C0 07 .....d3A-BN-IdhA
0000000060 1F 1E 68 66 00 CB 88 16-0E 00 66 81 3E 03 00 4E .....hf-E---f->-N
0000000070 54 46 53 75 15 B4 41 BB-AA 55 CD 13 72 0C 81 FB TFSu-Aa*UI-r-d
0000000080 55 AA 75 06 F7 C1 01 00-75 03 E9 DD 00 1E 83 EC U+u-A-u-eY-i
0000000090 18 68 1A 00 B4 48 8A 16-0E 00 8B F4 16 1F CD 13 .....h-H---O-i-
00000000a0 9F 83 C4 18 9E 58 1F 72-E1 3B 06 0B 00 75 DB A3 .....A-X-rá;-uDe
00000000b0 0F 00 C1 2E 0F 04 1E-5A 33 DB B9 00 20 2B C8 .....A---Z3Ü+eE
00000000c0 66 FF 06 11 00 03 16 0F-00 8E C2 FF 06 16 00 E8 fy-----Äy-e
00000000d0 4B 00 2B C8 77 EF B8 00-BB CD 1A 66 23 C0 75 2D K+EWi-;f#Au-
00000000e0 66 81 FB 54 43 50 41 75-24 81 F9 02 01 72 1E 16 f-üTCPaus-d-r-
00000000f0 68 07 BB 16 68 52 11 16-68 09 00 66 53 66 53 66 h-a-hR-h-;fSfSf
0000000100 55 16 16 16 68 B8 01 66-61 0E 07 CD 1A 33 C0 BF U---h,-fa-I-3Ä;
0000000110 0A 13 B9 F6 0C FC F3 AA-E9 FE 01 90 90 66 60 1E .....b-ü*ép---f-
0000000120 06 66 A1 11 00 66 03 06-1C 00 1E 66 68 00 00 00 .....f;-f---zh-
0000000130 00 66 50 06 53 68 01 00-68 10 00 B4 42 8A 16 0E .....fP-Sh-h-B-
0000000140 00 16 1F 8B F4 CD 13 66-59 5B 5A 66 59 66 59 1F .....öI-fYIZfYfY
0000000150 0F 82 16 00 66 FF 06 11-00 03 16 0F 00 8E C2 FF .....fy-----Äy
0000000160 0E 16 00 75 BC 07 1F 66-61 C3 A1 F6 01 E8 09 00 .....u-;-faÄ;ö-e-
0000000170 A1 FA 01 E8 03 00 F4 EB-FD 8B F0 AC 3C 00 74 09 .....ü-e-öey-ä-<-t-
0000000180 B4 0E BB 07 00 CD 10 EB-F2 C3 0D 0A 41 20 64 69 .....-i-eöÄ-A di
0000000190 73 6B 20 72 65 61 64 20-65 72 72 6F 72 20 6F 63 sk read error oc
00000001a0 63 75 72 72 65 64 00 0D-0A 42 4F 4F 54 4D 47 52 curred---BOOTMR
00000001b0 20 69 73 20 63 6F 6D 70-72 65 73 73 65 64 00 0D is compressed-
00000001c0 0A 50 72 65 73 73 20 43-74 72 6C 2B 41 6C 74 2B Press Ctrl+Alt+
00000001d0 44 65 6C 20 74 6F 20 72-65 73 74 61 72 74 0D 0A Del to restart-
00000001e0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
00000001f0 00 00 00 00 00 00 8A 01-A7 01 BF 01 00 00 55 AA .....$-;-U*
0000000200 07 00 42 00 4F 00 4F 00-54 00 4D 00 47 00 52 00 .....B-O-O-T-M-G-R-
0000000210 04 00 24 00 49 00 33 00-30 00 00 D4 00 00 00 24 .....-G-I-3-O-Ö---G
0000000220 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
0000000230 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
0000000240 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
0000000250 00 00 00 00 00 00 E9 C0-00 90 05 00 4E 00 54 00 .....-äÄ--N-T-
0000000260 4C 00 44 00 52 00 07 00-42 00 4F 00 4F 00 54 00 L-D-R-;-B-O-O-T-
0000000270 54 00 47 00 54 00 07 00-42 00 4F 00 4F 00 54 00 T-G-T-;-B-O-O-T-
```

```
0000000000 EB 52 90 4E 54 46 53 20-20 20 20 00 02 08 00 00 ER NTFS .....
0000000010 00 00 00 00 00 F8 00 00-3F 00 00 00 C8 0B 00 .....e-?g-E-
0000000020 00 00 00 00 80 00 80 00-47 92 3B 00 00 00 00 .....FC:
0000000030 00 00 0C 00 00 00 00 00-02 00 00 00 00 00 00 .....
0000000040 F6 00 00 00 01 00 00 00-20 3D C0 BE 88 C0 BE BC 0 .....ANAN
0000000050 00 00 00 00 FA 33 C0 8E-D0 BC 00 7C FB 68 C0 07 .....d3A-BN-IdhA
0000000060 1F 1E 68 66 00 CB 88 16-0E 00 66 81 3E 03 00 4E .....hf-E---f->-N
0000000070 54 46 53 75 15 B4 41 BB-AA 55 CD 13 72 0C 81 FB TFSu-Aa*UI-r-d
0000000080 55 AA 75 06 F7 C1 01 00-75 03 E9 DD 00 1E 83 EC U+u-A-u-eY-i
0000000090 18 68 1A 00 B4 48 8A 16-0E 00 8B F4 16 1F CD 13 .....h-H---O-i-
00000000a0 9F 83 C4 18 9E 58 1F 72-E1 3B 06 0B 00 75 DB A3 .....A-X-rá;-uDe
00000000b0 0F 00 C1 2E 0F 04 1E-5A 33 DB B9 00 20 2B C8 .....A---Z3Ü+eE
00000000c0 66 FF 06 11 00 03 16 0F-00 8E C2 FF 06 16 00 E8 fy-----Äy-e
00000000d0 4B 00 2B C8 77 EF B8 00-BB CD 1A 66 23 C0 75 2D K+EWi-;f#Au-
00000000e0 66 81 FB 54 43 50 41 75-24 81 F9 02 01 72 1E 16 f-üTCPaus-d-r-
00000000f0 68 07 BB 16 68 52 11 16-68 09 00 66 53 66 53 66 h-a-hR-h-;fSfSf
0000000100 55 16 16 16 68 B8 01 66-61 0E 07 CD 1A 33 C0 BF U---h,-fa-I-3Ä;
0000000110 0A 13 B9 F6 0C FC F3 AA-E9 FE 01 90 90 66 60 1E .....b-ü*ép---f-
0000000120 06 66 A1 11 00 66 03 06-1C 00 1E 66 68 00 00 00 .....f;-f---zh-
0000000130 00 66 50 06 53 68 01 00-68 10 00 B4 42 8A 16 0E .....fP-Sh-h-B-
0000000140 00 16 1F 8B F4 CD 13 66-59 5B 5A 66 59 66 59 1F .....öI-fYIZfYfY
0000000150 0F 82 16 00 66 FF 06 11-00 03 16 0F 00 8E C2 FF .....fy-----Äy
0000000160 0E 16 00 75 BC 07 1F 66-61 C3 A1 F6 01 E8 09 00 .....u-;-faÄ;ö-e-
0000000170 A1 FA 01 E8 03 00 F4 EB-FD 8B F0 AC 3C 00 74 09 .....ü-e-öey-ä-<-t-
0000000180 B4 0E BB 07 00 CD 10 EB-F2 C3 0D 0A 41 20 64 69 .....-i-eöÄ-A di
0000000190 73 6B 20 72 65 61 64 20-65 72 72 6F 72 20 6F 63 sk read error oc
00000001a0 63 75 72 72 65 64 00 0D-0A 42 4F 4F 54 4D 47 52 curred---BOOTMR
00000001b0 20 69 73 20 63 6F 6D 70-72 65 73 73 65 64 00 0D is compressed-
00000001c0 0A 50 72 65 73 73 20 43-74 72 6C 2B 41 6C 74 2B Press Ctrl+Alt+
00000001d0 44 65 6C 20 74 6F 20 72-65 73 74 61 72 74 0D 0A Del to restart-
00000001e0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
00000001f0 00 00 00 00 00 00 8A 01-A7 01 BF 01 00 00 55 AA .....$-;-U*
0000000200 07 00 42 00 4F 00 4F 00-54 00 4D 00 47 00 52 00 .....B-O-O-T-M-G-R-
0000000210 04 00 24 00 49 00 33 00-30 00 00 D4 00 00 00 24 .....-G-I-3-O-Ö---G
0000000220 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
0000000230 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
0000000240 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
0000000250 00 00 00 00 00 00 E9 C0-00 90 05 00 4E 00 54 00 .....-äÄ--N-T-
0000000260 4C 00 44 00 52 00 07 00-42 00 4F 00 4F 00 54 00 L-D-R-;-B-O-O-T-
0000000270 54 00 47 00 54 00 07 00-42 00 4F 00 4F 00 54 00 T-G-T-;-B-O-O-T-
```

0	46	49	4C	45	30	00	03	00-DE	50	04	E3	47	00	00	00	FILE0...	PP	AG...							
0	0F	00	02	00	38	00	01	00-E0	02	00	00	00	00	04	00	00	...	8...	à...						
0	00	00	00	00	00	00	00	00-0F	00	00	00	D4	AB	03	00	00	...	...	Ô«...						
0	04	00	B1	A3	00	00	00	00-10	00	00	00	60	00	00	00	00	...	±f...	'...						
0	00	00	00	00	00	00	00	00-48	00	00	00	18	00	00	00	00	...	...	H...						
0	DE	EE	C9	44	AF	A4	DB	01-BD	C8	40	46	AF	A4	DB	01	01	BiED~u0	~E0F~u0							
0	1C	38	1C	5B	90	A5	DB	01-83	75	4B	59	90	A5	DB	01	01	8	[~u0	~uKY~u0						
0	20	00	04	00	00	00	00	00-00	00	00	00	00	00	00	00	00	...	...	...						
0	00	00	00	00	77	75	00	00-00	00	00	00	00	00	00	00	00	...	wu...	...						
0	30	B8	2C	1F	08	00	00	00-30	00	00	00	78	00	00	00	00	0,	...	0...x...						
0	00	00	00	00	00	00	08	00-5A	00	00	00	18	00	01	00	00	...	...	Z...						
0	D1	89	03	00	00	00	11	00-DE	EE	C9	44	AF	A4	DB	01	01	N~	...	BiED~u0						
0	76	17	CF	44	AF	A4	DB	01-76	17	CF	44	AF	A4	DB	01	01	v	~ID~u0	v	~ID~u0					
0	76	17	CF	44	AF	A4	DB	01-00	70	0C	00	00	00	00	00	00	v	~ID~u0	..p	...					
0	28	6E	0C	00	00	00	00	00-20	00	00	00	00	00	00	00	00	(n	...	...						
0	0C	02	4A	00	47	00	43	00-53	00	33	00	34	00	7E	00	00	..J	G	C	S	3	4	~		
0	31	00	2E	00	5A	00	49	00-50	00	00	00	00	00	00	00	00	1	..	Z	I	P	...	...		
0	30	00	00	00	78	00	00	00-00	00	00	00	00	00	00	00	07	0	...	x	...	...	...	...		
0	60	00	00	00	18	00	01	00-D1	89	03	00	00	00	00	11	00	...	...	N~	...	...	...	...		
0	DE	EE	C9	44	AF	A4	DB	01-76	17	CF	44	AF	A4	DB	01	01	BiED~u0	v	~ID~u0	...	...	...	...	...	
0	76	17	CF	44	AF	A4	DB	01-76	17	CF	44	AF	A4	DB	01	01	v	~ID~u0	v	~ID~u0	...	...	...	...	
0	00	70	0C	00	00	00	00	00-28	6E	0C	00	00	00	00	00	00	..p	...	(n	...	...	...	...		
0	20	00	00	00	00	00	00	00-0F	01	6A	00	67	00	63	00	00	...	...	j	g	c	...	...		
0	73	00	33	00	34	00	30	00-20	00	28	00	32	00	29	00	00	s	3	4	0	..	..	(	2	..)
0	2E	00	7A	00	69	00	70	00-40	00	00	00	28	00	00	00	00	..z	i	p	@	...	(	...	...	
0	00	00	00	00	00	00	00	00-10	00	00	00	18	00	00	00	00	...	...	...	...	...	...	...	...	
0	A1	32	E8	69	9D	10	F0																		

af41e0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
af4ff0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 05 00	
af5000	46 49 4C 45 30 00 03 00-DE 50 04 E3 47 00 00 00	FILE0...BP-ãG...
af5010	0F 00 02 00 38 00 01 00-E0 02 00 00 00 04 00 00	...8...â...
af5020	00 00 00 00 00 00 00 00 00-0F 00 00 00 D4 AB 03 00	Ô«...
af5030	04 00 B1 A3 00 00 00 00-10 00 00 00 60 00 00 00	...f.....
af5040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	H.....
af5050	DE EE C9 44 AF A4 DB 01-BD C8 40 46 AF A4 DB 01	piÉD¬û·-z@F¬û·
af5060	1C 38 1C 5B 90 A5 DB 01-83 75 4B 59 90 A5 DB 01	·8·[·¬û··uKY·¬û·
af5070	20 00 04 00 00 00 00 00-00 00 00 00 00 00 00 00	
af5080	00 00 00 00 77 75 00 00-00 00 00 00 00 00 00 00	...wu.....
af5090	30 B8 2C 1F 08 00 00 00-30 00 00 00 78 00 00 00	0,,...0...x...
af50a0	00 00 00 00 00 00 08 00-5A 00 00 00 18 00 01 00	Z.....
af50b0	D1 89 03 00 00 00 11 00-DE EE C9 44 AF A4 DB 01	Ñ.....piÉD¬û·
af50c0	76 17 CF 44 AF A4 DB 01-76 17 CF 44 AF A4 DB 01	v·iD¬û·v·iD¬û·
af50d0	76 17 CF 44 AF A4 DB 01-00 70 0C 00 00 00 00 00	v·iD¬û·p.....
af50e0	28 6E 0C 00 00 00 00 00-20 00 00 00 00 00 00 00	(n.....
af50f0	0C 02 4A 00 47 00 43 00-53 00 33 00 34 00 7E 00	..J·G·C·S·3·4·~·
af5100	31 00 2E 00 5A 00 49 00-50 00 00 00 00 00 00 00	l·.·Z·I·P.....
af5110	30 00 00 00 78 00 00 00-00 00 00 00 00 00 07 00	0...x.....
af5120	60 00 00 00 18 00 01 00-D1 89 03 00 00 00 11 00	`.....Ñ.....
af5130	DE EE C9 44 AF A4 DB 01-76 17 CF 44 AF A4 DB 01	piÉD¬û·v·iD¬û·
af5140	76 17 CF 44 AF A4 DB 01-76 17 CF 44 AF A4 DB 01	v·iD¬û·v·iD¬û·
af5150	00 70 0C 00 00 00 00 00-28 6E 0C 00 00 00 00 00	·p.....(n.....
af5160	20 00 00 00 00 00 00 00-0F 01 6A 00 67 00 63 00	j·g·c·
af5170	73 00 33 00 34 00 30 00-20 00 28 00 32 00 29 00	s·3·4·0·.·(·2·)·
af5180	2E 00 7A 00 69 00 70 00-40 00 00 00 28 00 00 00	..z·i·p·@...·(...
af5190	00 00 00 00 00 00 0E 00-10 00 00 00 18 00 00 00	
af51a0	A1 32 E8 69 9D 10 F0 11-8B 5A 10 3D 1C 03 45 F2	;2èi··ð··Z·=··Eò
af51b0	80 00 00 00 58 00 00 00-01 00 00 00 00 00 04 00	...X.....
af51c0	00 00 00 00 00 00 00 00-C6 00 00 00 00 00 00 00	E.....
af51d0	40 00 00 00 00 00 00 00-00 70 0C 00 00 00 00 00	@.....p.....
af51e0	28 6E 0C 00 00 00 00 00-28 6E 0C 00 00 00 00 00	(n.....(n.....
af51f0	41 09 4C DD F8 00 41 7F-86 49 62 01 41 3F 04 00	A·LÝø·A··Ib·A?·
af5200	C0 FF 00 00 00 00 00 00-80 00 00 00 68 00 00 00	âh.....h...

File-created Timestamp

File-Modified Timestamp

File-Modified Timestamp

Hex Value Interpreter			File List			
Type	Size	Value	Name	Size	Type	Date Modified
signed integer	1-8	133,881,686,642,444,477	DumpStack.log.tmp	8	Regular File	4/4/2025 9:00:46 AM
unsigned integer	1-8	133,881,686,642,444,477	DumpStack.log	8	Regular File	10/13/2024 9:00:49 A
FILETIME (UTC)	8	4/3/2025 3:44:24 PM	D82FA9F7F0EF	1	Regular File	1/4/2024 4:21:03 PM
FILETIME (local)	8	4/3/2025 11:44:24 AM	852EC3DCD415	1	Regular File	9/14/2021 5:53:33 PM
DOS_date	2	.	\$Volume	0	Regular File	12/7/2021 2:07:20 PM
DOS_time	2	.	\$UpCase	128	Regular File	12/7/2021 2:07:20 PM
time_t (UTC)	4	.	\$TXF_DATA	1	NTFS Logged ...	4/4/2025 12:42:10 PM
time_t (local)	4	.	\$Secure	1	Regular File	12/7/2021 2:07:20 PM
			\$MFTMirr	4	Regular File	12/7/2021 2:07:20 PM
			\$MFT	1,499,392	Regular File	12/7/2021 2:07:20 PM
			\$LogFile	65,536	Regular File	12/7/2021 2:07:20 PM
			\$I30	8	NTFS Index All...	4/4/2025 12:42:10 PM
			\$Boot	8	Regular File	12/7/2021 2:07:20 PM
			0eaf4f80	00 00 00 00 00 00 00 00-00	00 00 00 00 00 00 00 00	
			0eaf4f90	00 00 00 00 00 00 00 00-00	00 00 00 00 00 00 00 00	
			0eaf4fa0	00 00 00 00 00 00 00 00-00	00 00 00 00 00 00 00 00	
			0eaf4fb0	00 00 00 00 00 00 00 00-00	00 00 00 00 00 00 00 00	
			0eaf4fc0	00 00 00 00 00 00 00 00-00	00 00 00 00 00 00 00 00	
			0eaf4fd0	00 00 00 00 00 00 00 00-00	00 00 00 00 00 00 00 00	
			0eaf4fe0	00 00 00 00 00 00 00 00-00	00 00 00 00 00 00 00 00	
			0eaf4ff0	00 00 00 00 00 00 00 00-00	00 00 00 00 00 00 05 00	
			0eaf5000	46 49 4C 45 30 00 03 00-DE	50 04 E3 47 00 00 00 00	FILE0...BP·AG·...
			0eaf5010	0F 00 02 00 38 00 01 00-E0	02 00 00 00 04 00 00 00	...·8...à...·
			0eaf5020	00 00 00 00 00 00 00 00-0F	00 00 00 D4 AB 03 00 00	...·...·Ö«...
			0eaf5030	04 00 B1 A3 00 00 00 00-10	00 00 00 60 00 00 00 00	...±E...·...
			0eaf5040	00 00 00 00 00 00 00 00-48	00 00 00 18 00 00 00 00	...·...H...·
			0eaf5050	DE EE C9 44 AF A4 DB 01-BD	C8 40 46 AF A4 DB 01	piÉD·hÜ·+E@F·hÜ·
			0eaf5060	1C 38 1C 5B 90 A5 DB 01-83	75 4B 59 90 A5 DB 01	·8·[·YÜ··uKY·YÜ·
			0eaf5070	20 00 04 00 00 00 00 00-00	00 00 00 00 00 00 00 00	...·wu...·
			0eaf5080	00 00 00 00 77 75 00 00-00	00 00 00 00 00 00 00 00	...·...·
			0eaf5090	30 B8 2C 1F 08 00 00 00-30	00 00 00 78 00 00 00 00	0,,...·0...·x...·
			0eaf50a0	00 00 00 00 00 00 08 00-5A	00 00 00 18 00 01 00 00	...·...·Z...·
			0eaf50b0	D1 89 03 00 00 00 11 00-DE	EE C9 44 AF A4 DB 01	N...·piÉD·hÜ·
			0eaf50c0	76 17 CF 44 AF A4 DB 01-76	17 CF 44 AF A4 DB 01	v·iD·hÜ·v·iD·hÜ·
			0eaf50d0	76 17 CF 44 AF A4 DB 01-00	70 0C 00 00 00 00 00 00	v·iD·hÜ··p...·
			0eaf50e0	28 6E 0C 00 00 00 00 00-20	00 00 00 00 00 00 00 00	(n...·...·
			0eaf50f0	0C 02 4A 00 47 00 43 00-53	00 33 00 34 00 7E 00	·J·G·C·S·3·4·w·
			0eaf5100	31 00 2E 00 5A 00 49 00-50	00 00 00 00 00 00 00 00	1·,·Z·I·P...·
			0eaf5110	30 00 00 00 78 00 00 00-00	00 00 00 00 00 07 00	0...·x...·
			0eaf5120	60 00 00 00 18 00 01 00-D1	89 03 00 00 00 11 00	`...·...·N...·

MFT Record-Changed Timestamp

[illegible]

File-Accessed Timestamp

Hex Value Interpreter

Type	Size	Value
signed integer	1-8	133,882,653,329,552,771
unsigned integer	1-8	133,882,653,329,552,771
FILETIME (UTC)	8	4/4/2025 6:35:32 PM
FILETIME (local)	8	4/4/2025 2:35:32 PM
DOS date	2	.
DOS time	2	.
time_t (UTC)	4	.
time_t (local)	4	.

File List

Name	Size	Type	Date Modified
DumpStack.log.tmp	8	Regular File	4/4/2025 9:00:46 AM
DumpStack.log	8	Regular File	10/13/2024 9:00:49 AM
D82FA9F7F0EF	1	Regular File	1/4/2024 4:21:03 PM
852EC3DCD415	1	Regular File	9/14/2021 5:53:33 PM
\$Volume	0	Regular File	12/7/2021 2:07:20 PM
\$UpCase	128	Regular File	12/7/2021 2:07:20 PM
\$TXF_DATA	1	NTFS Logged ...	4/4/2025 12:42:10 PM
\$Secure	1	Regular File	12/7/2021 2:07:20 PM
\$MFTMirr	4	Regular File	12/7/2021 2:07:20 PM
\$MFT	1,499,392	Regular File	12/7/2021 2:07:20 PM
\$LogFile	65,536	Regular File	12/7/2021 2:07:20 PM
\$I30	8	NTFS Index All...	4/4/2025 12:42:10 PM
\$Boot	8	Regular File	12/7/2021 2:07:20 PM

0eaf4f80

00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00

.....

0eaf4f90

00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00

.....

0eaf4fa0

00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00

.....

0eaf4fb0

00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00

.....

0eaf4fc0

00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00

.....

0eaf4fd0

00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00

.....

0eaf4fe0

00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00

.....

0eaf4ff0

00 00 00 00 00 00 00 00-00 00 00 00 00 05 00 00

.....

0eaf5000

46 49 4C 45 30 00 03 00-DE 50 04 E3 47 00 00 00

FILE0...FP-AG...

0eaf5010

0F 00 02 00 38 00 01 00-E0 02 00 00 00 04 00 00

...8...à...00 00

0eaf5020

00 00 00 00 00 00 00 00-0F 00 00 00 D4 AB 03 00

.....Öκ...

0eaf5030

04 00 B1 A3 00 00 00 00-10 00 00 00 60 00 00 00

...±f.....

0eaf5040

00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00

.....H.....

0eaf5050

DE EE C9 44 AF A4 DB 01-BD C8 40 46 AF A4 DB 01

BiEDmÛ·Aë@FmÛ·

0eaf5060

1C 38 1C 5B 90 A5 DB 01-83 75 4B 59 90 A5 DB 01

·8·[·YÜ··uKY·YÜ·

0eaf5070

20 00 04 00 00 00 00 00-00 00 00 00 00 00 00 00

.....

0eaf5080

00 00 00 00 77 75 00 00-00 00 00 00 00 00 00 00

...wu.....

0eaf5090

30 B8 2C 1F 08 00 00 00-30 00 00 00 78 00 00 00

0...·...0...x...

0eaf50a0

00 00 00 00 00 00 08 00-5A 00 00 00 18 00 01 00

.....Z.....

0eaf50b0

D1 89 03 00 00 00 11 00-DE EE C9 44 AF A4 DB 01

Ñ.....BiEDmÛ·

0eaf50c0

76 17 CF 44 AF A4 DB 01-76 17 CF 44 AF A4 DB 01

v·IDmÛ·v·IDmÛ·

0eaf50d0

76 17 CF 44 AF A4 DB 01-00 70 0C 00 00 00 00 00

v·IDmÛ··p.....

0eaf50e0

28 6E 0C 00 00 00 00 00-20 00 00 00 00 00 00 00

(n.....

0eaf50f0

0C 02 4A 00 47 00 43 00-53 00 33 00 34 00 7E 00

·J·G·C·S·3·4·w·

0eaf5100

31 00 2E 00 5A 00 49 00-50 00 00 00 00 00 00 00

1...Z·I·P.....

0eaf5110

30 00 00 00 78 00 00 00-00 00 00 00 00 07 00 00

0...x.....

0eaf5120

60 00 00 00 18 00 01 00-D1 89 03 00 00 00 11 00

·.....Ñ.....

0eaf5130

DE EE C9 44 AF A4 DB 01-76 17 CF 44 AF A4 DB 01

BiEDmÛ·v·IDmÛ·

0eaf5140

76 17 CF 44 AF A4 DB 01-76 17 CF 44 AF A4 DB 01

v·IDmÛ·v·IDmÛ·

0eaf5150

00 70 0C 00 00 00 00 00-28 6E 0C 00 00 00 00 00

·p.....(n.....

0eaf5160

20 00 00 00 00 00 00 00-0F 01 6A 00 67 00 63 00

.....j·g·c·

0eaf5170

73 00 33 00 34 00 30 00-20 00 28 00 32 00 29 00

s·3·4·0··(·2·)·

0eaf5180

2E 00 7A 00 69 00 70 00-40 00 00 00 28 00 00 00

·z·i·p·@·(·(·

0eaf5190

00 00 00 00 00 00 0E 00-10 00 00 00 18 00 00 00

.....

0eaf51a0

A1 32 E8 69 9D 10 F0 11-8B 5A 10 3D 1C 03 45 F2

j2èi··8··Z·=··Eò

DOS and Windows filename

DOS filename: jgcs340 (2).zip

0eaf5160	20 00 00 00 00 00 00 00 00-0F 01 6A 00 67 00 63 00	j·g·c·
0eaf5170	73 00 33 00 34 00 30 00-20 00 28 00 32 00 29 00	s·3·4·0· ·(-2·)·
0eaf5180	2E 00 7A 00 69 00 70 00-40 00 00 00 28 00 00 00	·z·i·p·@···(···
0eaf5190	00 00 00 00 00 00 00 00 0E 00-10 00 00 00 18 00 00 00	
0eaf51a0	A1 32 E8 69 9D 10 F0 11-8B 5A 10 3D 1C 03 45 F2	¡2èi·-ð·-Z·=-·Eò
0eaf51b0	80 00 00 00 00 58 00 00 00-01 00 00 00 00 00 04 00	X.....
0eaf51c0	00 00 00 00 00 00 00 00 00-C6 00 00 00 00 00 00 00	E.....
0eaf51d0	40 00 00 00 00 00 00 00 00-00 70 0C 00 00 00 00 00	@.....p.....
0eaf51e0	28 6E 0C 00 00 00 00 00 00-28 6E 0C 00 00 00 00 00	(n.....(n.....
0eaf51f0	41 09 4C DD F8 00 41 7F-86 49 62 01 41 3F 04 00	A·LÝø·A·-Ib·A?·
0eaf5200	C0 FE 00 00 00 00 00 00 00-80 00 00 00 68 00 00 00	Àp.....h···
0eaf5210	01 0F 40 00 00 00 00 0D 00-00 00 00 00 00 00 00 00	·@.....
0eaf5220	00 00 00 00 00 00 00 00 00-60 00 00 00 00 00 00 00	
0eaf5230	00 10 00 00 00 00 00 00 00-96 02 00 00 00 00 00 00	
0eaf5240	96 02 00 00 00 00 00 00 00-5A 00 6F 00 6E 00 65 00	Z·o·n·e·
0eaf5250	2E 00 49 00 64 00 65 00-6E 00 74 00 69 00 66 00	·I·d·e·n·t·i·f·
0eaf5260	69 00 65 00 72 00 00 00-31 01 20 DA 25 00 00 00	i·e·r···l· Ú\$···
0eaf5270	D0 00 00 00 20 00 00 00 00-00 00 00 00 00 00 09 00	Đ···
0eaf5280	08 00 00 00 18 00 00 00-83 03 00 00 8C 03 00 00	
0eaf5290	E0 00 00 00 48 00 00 00-01 00 00 00 00 00 0B 00	à···H.....
0eaf52a0	00 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	
0eaf52b0	40 00 00 00 00 00 00 00 00-00 10 00 00 00 00 00 00	@.....

Data run :

1: 41 09 4C DD F8 00

2: 41 7F 86 49 62 01

3: 41 3F 04 00 C0 FE

4. For the file in problem 3, locate all cluster addresses (if more than one). Verify the following:

Starting sector (using WinHex)

First sector of any additional clusters (using WinHex)

File size (using file properties)

